

| Model       | Computation | Method      | Mean time (512 taxa) | log-log slope |
|-------------|-------------|-------------|----------------------|---------------|
| JC          | Likelihood  | TreeFlow    | 4.16                 | 1.19          |
|             |             | bito/BEAGLE | 0.07                 | 1.07          |
|             |             | JAX         | 23.65                | 1.01          |
|             | Gradients   | TreeFlow    | 11.98                | 1.36          |
|             |             | bito/BEAGLE | 0.43                 | 1.25          |
|             |             | JAX         | 312.68               | 1.11          |
| GTR/Weibull | Likelihood  | TreeFlow    | 10.52                | 1.16          |
|             |             | bito/BEAGLE | 0.26                 | 1.23          |
|             |             | JAX         | 65.39                | 1.27          |
|             | Gradients   | TreeFlow    | <b>41.00</b>         | <b>1.27</b>   |
|             |             | bito/BEAGLE | <b>6.84</b>          | <b>1.31</b>   |
|             |             | JAX         | <b>533.91</b>        | <b>1.17</b>   |

Table 1: Results of phylogenetic likelihood benchmark. Times for gradients for the GTR/Weibull are highlighted as they are the most relevant computation for gradient-based inference on real data.